

Muhammad Umar Khan

Signal Processing | Multimodal Learning | AI in Healthcare | Neurophysiological Decoding | Biomedical Computer Vision
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Location: Canberra, ACT, Australia
Research Metrics: 100+ publications | 1995 citations | h-index 25 | i10-index 57
Profiles: [Google Scholar](#) | [ResearchGate](#) | [ORCID](#)
Memberships: IEEE | ACM | ACS

Targeted Research Profile

- Research scientist with an electronics engineering background and 10+ years of experience in biomedical signal processing, multimodal AI, machine/deep learning, physiological sensing, and embedded intelligent systems.
- Final-year PhD candidate developing multimodal AI methods for objective health monitoring using brain and body signals.
- Strong alignment with neurotechnology and BCI research through work on neurophysiological signal modelling and decoding, temporal representation learning, multimodal data fusion, subject-independent classification, and wearable biosensing.
- Published 100+ peer-reviewed articles, including work in *Information Fusion*, *ACM Computing Surveys*, *Biomedical Signal Processing and Control*, and *Computers in Biology and Medicine*.

Core Technical Experience

<u>Neurophysiological Signals</u>	EEG, fNIRS, EDA, BVP/PPG, ECG, EMG, PCG, wearable biosignals, physiological time-series modelling, neurophysiological decoding
<u>Machine/Deep Learning</u>	CNNs, LSTMs, Transformers, multimodal learning, representation learning, classification, feature learning, lightweight model design
<u>Signal Processing</u>	Time-domain, frequency-domain and time-frequency analysis, filtering, feature engineering, temporal representation, signal fusion
<u>Data Collection Systems</u>	BIOPAC MP36, EmotiBit, Artinis fNIRS System, and Plux Biosignals systems; physiological data acquisition; experimental protocol and setup; sensor placement and handling
<u>Embedded and Real-Time AI</u>	Raspberry Pi, STM32, embedded signal processing, real-time processing, sensor-based systems, technical testing and documentation
<u>Medical Imaging and Vision</u>	Medical image analysis, video-based assessment, biomedical computer vision
<u>Programming and Tools</u>	Python, MATLAB, Simulink, C/C++, PyTorch, TensorFlow, Anaconda, Git, LaTeX, technical writing

Professional Experience

<u>ICC Research Associate</u>	2026–Present
<i>Innovation Central Canberra, University of Canberra, ACT, Australia</i>	
• Design, implement, test, and document real-time digital signal processing solutions for industry-linked embedded systems R&D projects.	
<u>Doctoral Researcher</u>	2023–Present
<i>Centre for Intelligent Computing and Systems, University of Canberra, ACT, Australia</i>	
• Developing novel multimodal AI methods for health monitoring using neurophysiological brain and body signals, with a focus on temporal representation learning, multimodal fusion, and subject-independent modelling. • Conducting research across biomedical signal processing, affective computing, human-centred AI, wearable health technologies, and real-time physiological sensing.	
<u>Team Lead, Signal/Image Processing</u>	2016–2024
<i>SEED Research Group, UET Taxila, Pakistan</i>	

- Co-founded and led signal/image processing research activities, supervising collaborative projects in biomedical signal processing, medical AI, machine learning, computer-aided diagnosis, and embedded intelligent systems.

Lab Engineer

2014–2024

Department of Electronics Engineering, UET Taxila, Pakistan

- Conducted laboratory teaching and technical instruction in signal/image processing, programming, control and communications courses.

Casual Academic

2024–Present

Faculty of Science and Technology, University of Canberra, ACT, Australia

- Delivered teaching support across computing, software development, mobile technologies, and data analysis units, including tutorials, laboratory sessions, assessment marking, and student support.

Selected Research Contributions

- Developed multimodal physiological signal processing methods for health assessment using EDA, BVP/PPG, fNIRS, EEG, ECG, EMG, and PCG signals.
- Designed machine/deep learning pipelines for subject-independent classification, multimodal fusion, and temporal feature representation in physiological time-series data.
- Experienced in multimodal physiological data collection using systems from BIOPAC, EmotiBit, and Plux Biosignals, covering EDA, BVP/PPG, ECG, EMG, and related wearable biosignals.
- Built research systems involving wearable biosensing, embedded processing, real-time signal analysis, and computer-aided diagnostic modelling.
- Led and supervised applied research projects across biosignal-based cardiac diagnosis, biometric authentication, respiratory sound analysis, neuromuscular disorder assessment, and motor fault detection.

Education

PhD | Biomedical Signal Processing and AI in Healthcare

2023–Present

University of Canberra, ACT, Australia

- Thesis: Multimodal Signal Processing based Identification of Human Pain.
- PhD Supervisory Panel: [Dr Raul Fernandez Rojas](#), [Prof. Girija Chetty](#), and [Prof. Roland Goecke](#); thesis submission planned for August 2026.

BS | Electronics Engineering

2008–2012

COMSATS Institute of Information Technology, Pakistan

- Final Year Project: Detection and Classification of Heart Murmurs for Cardiac Disorders.
- CGPA: 3.53/4.00.

Selected Publications

1. **Khan, M. U.**, Chetty, G., Gkikas, S., Tsiknakis, M., Goecke, R., & Fernandez-Rojas, R. (2026). GIAFormer: A Gradient-Infused Attention and Transformer for Pain Assessment with EDA–fNIRS Fusion. *Information Fusion*, 104173.
2. **Khan, M. U.**, Hirachan, N., Joseph, C., Murtagh, L., Chetty, G., Goecke, R., & Fernandez-Rojas, R. (2026). Pain intensity estimation via multimodal fusion: Leveraging ternary textures of derivatives in EDA and PPG signals. *Biomedical Signal Processing and Control*, 112, 108532.
3. **Khan, M. U.**, Chetty, G., Goecke, R., & Fernandez-Rojas, R. (2025). A Systematic Review of Multimodal Signal Fusion for Acute Pain Assessment Systems. *ACM Computing Surveys*, 58(2), 1–35.
4. **Khan, M. U.**, Aziz, S., Murtagh, L., Chetty, G., Goecke, R., & Rojas, R. F. (2025). Empirically Transformed Energy Patterns: A novel approach for capturing fNIRS signal dynamics in pain assessment. *Computers in Biology and Medicine*, 192, 110300.
5. **Khan, M. U.**, Sousani, M., Hirachan, N., Joseph, C., Ghahramani, M., Chetty, G., Goecke, R., & Fernandez-Rojas, R. (2024). Multilevel Pain Assessment with Functional Near-Infrared Spectroscopy. *Sensors*, 24(2), 458.

Complete publication list available through [Google Scholar](#).

Professional Service and Research Community Engagement

<u>Program Committee</u>	AI4Pain Challenge 2026; AI4Pain Challenge 2025; 14th International Conference on Affective Computing and Intelligent Interaction (ACII 2026), Puebla, Mexico
<u>Session Chair</u>	IEEE International Conference on Electrical, Communication, and Computer Engineering (ICECCE 2020), Istanbul, Turkey
<u>Reviewer</u>	Biomedical Signal Processing and Control; Engineering Applications of Artificial Intelligence; Expert Systems with Applications; Journal of Medical Systems; IEEE Neural Engineering Conference (NER 2025), San Diego, USA; 14th International Conference on Affective Computing and Intelligent Interaction (ACII 2026), Puebla, Mexico
<u>Conference Volunteer</u>	ACII 2025, Canberra, Australia; 27th ACM International Conference on Multimodal Interaction (ICMI 2025), Canberra, Australia

Selected Teaching Experience

University of Canberra, Australia (2024–Present): Data Capture and Preparation; Software Technology 2; Mobile Technologies; Web Design and Programming; Introduction to Computer Engineering; AR/VR for Data Analysis and Communication.

University of Engineering and Technology Taxila, Pakistan (2014–2024): Signals and Systems; Digital Signal Processing; Digital Image Processing; Biomedical Signal Analysis; Computer Programming; Analogue and Digital Communications; Control Systems; Artificial Intelligence; Medical Imaging; Biomedical Instrumentation.

Awards and Honours

- W. J. Weeden Postgraduate Research Stipend Scholarship and Australian Government Research Training Program (RTP) Scholarship for PhD studies, University of Canberra, Australia.
- National ICT R&D Scholarship for BS in Electronics Engineering, Government of Pakistan.
- First Prize, BEE-HIPE Project Exhibition, COMSATS Institute of Information Technology, 2011.
- Third Prize, Open House 2012, awarded by the Pakistan Engineering Council.

References

1. Dr Raul Fernandez Rojas

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University of Canberra, Australia
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3. Prof. Roland Goecke

Professor; Head of School, Systems & Computing
UNSW Canberra, Australia
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2. Prof. Girija Chetty

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Additional Referees

Available upon request